Andrea Wan

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UBC Integrated Engineering student with experience in mechanical design, mechatronic prototyping, and substation engineering. Skilled in **SolidWorks**, **3D printing**, microcontroller integration, and engineering drawing review.

PROFESSIONAL EXPERIENCE

Engineering Intern - Substation | FortisBC **May 2026 – Present**
Trail, BC

- **Field Verification:** Verify field wiring for 10+ transformer installations against 50+ drawings and identify discrepancies.
- **Engineering Calculations:** Support 15+ operating-limit and calculation packages through equipment data compilation and technical verification.
- **Materials & Documentation:** Standardize 20+ material lists in **SAP** and **Excel** and maintain 30+ technical documents for procurement and field implementation.
- **Lighting Design:** Prepare lighting upgrade packages for 3 substations using as-built drawings; simplify work summaries to reduce administrative workload by 25%.
- **Transformer Cooling:** Calculated auxiliary loads for 4 additional cooling fans (2 per stage; 12 total), updated drawings, and completed the Issued for Construction (**IFC**) package.
- **Data Automation:** Automate **SAP** extraction and **Excel** processing; build dashboards to track assets, materials, and engineering activities.

Engineering Intern **March 2024 – June 2024**
China Penghui Environmental Protection Engineering Ltd. China

- **Equipment Reliability:** Supported monitoring station maintenance and mechanical inspections, helping sustain 95% sensor uptime.
- **Data & Documentation:** Recorded data for 10+ wastewater facilities, improving reporting accuracy by 12%; standardized site records and equipment logs to improve audit efficiency by 25%.
- **Project Support:** Supported 3 municipal wastewater upgrades, meeting technical milestones 1 week early; monitored station output against environmental requirements.

EDUCATION

Bachelor of Applied Science in Integrated Engineering **2024 - 2029 (Expected)**
University of British Columbia | CGPA: 82.6% (A-) Vancouver, BC

Bachelor of Science in Mathematics, Operational Research, Statistics, and Economics **2022 – 2023**
University of Warwick Coventry, UK

PROJECTS

Sound-Responsive Automatic Page Turner | UBC **2026**

- **Mechatronic Integration:** Developed an **ESP32**-based system using MIDI and audio sensing, achieving a 98% successful page-turn rate.
- **Signal Processing:** Implemented **particle filtering**, moving-average filtering, and frequency-feature extraction, reducing ambient noise interference by 40%.
- **Mechanical Prototyping:** Designed 12 components in **SolidWorks** and **3D-printed** them, reducing device weight by 15%.

Line-Following Autonomous Robot | UBC **2025**

- **Navigation & Motor Control:** Programmed **Arduino** and IR-based line following with 90% tracking accuracy; tuned differential motor control to improve cornering stability by 25%.
- **Mechanical & PCB Design:** Designed a **SolidWorks** obstacle-detection mechanism and custom motor **PCBs**, achieving 100% detection reliability and reducing electrical interference by 30%.

Quadruped Robot | Personal Project **June 2024**

- **Hardware Integration:** Modified an existing **Arduino** design with 4 servos, PLA parts, and ultrasonic sensing; **3D-printed** a custom cat-shaped sensor housing and tuned obstacle-triggered stopping, achieving near-100% test success.

TECHNICAL SKILLS

Mechanical: SolidWorks (**CSWA**), 3D CAD, 2D Drawings, **3D Printing**, Tolerance Stack-Up Analysis, Materials, Basic Machining
Electrical: Arduino Uno, **ESP32**, Circuit Soldering & Prototyping, Substation Design, Power Distribution Calculations, Schematics
Software: C++, C#, Java, **Excel** (Advanced Data Modeling), VBA, Bluebeam Revu, **SAP**
Project Delivery: Planning, Scheduling, Documentation, Coordination, Risk Assessment, Progress Tracking, Engineering Change Management, Stakeholder Communication